CSC 3320: System-Level Programming

System Programming and Introduction to UNIX

Spring 2026

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Department of Computer Science, GSU

Last class agenda

- Introduced syllabus and course policy
- Introduced system-level programming
- Learned how to *log in* **snowball.cs.gsu.edu** using **ssh** client
- Learned how to use *sftp, put, get* to upload and download files to/from the server.

Chapter 1
What is UNIX?

Graham Glass and King Ables,
UNIX for Programmers and Users,
Third Edition, Pearson Prentice Hall, 2003.

Original Notes by Raj Sunderraman

Converted to presentation and updated by Michael Weeks

*Many slides are created/modified by the instructor

Computer Systems

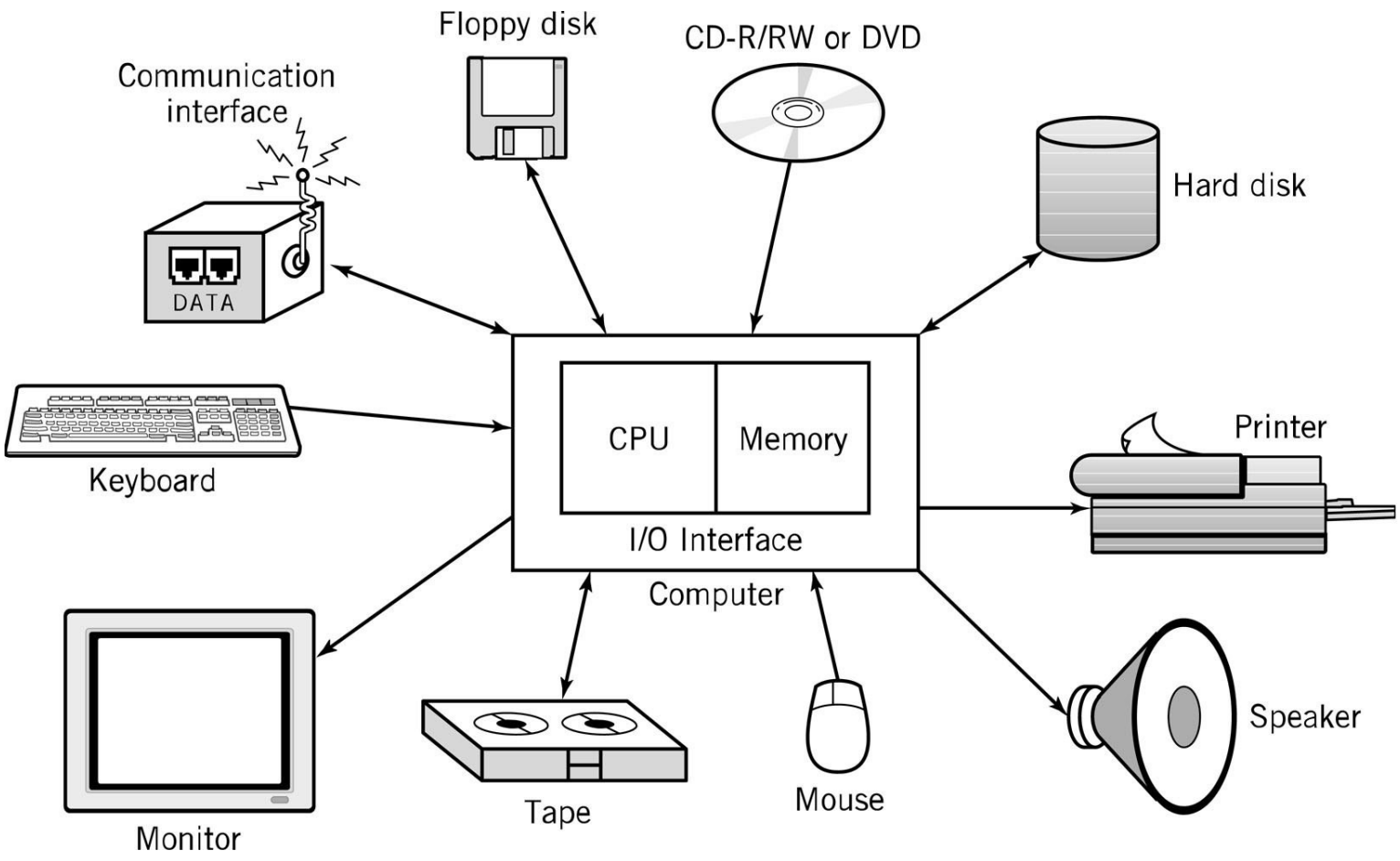
- **Computer System : Hardware + Software**

- **Hardware: CPU, Memory (RAM/ROM), Disk drives, CD-ROM drives, Monitor, Graphics card, Keyboard, Mouse, Printer, Tape drive, Modem, Ethernet interface, Other peripherals.**

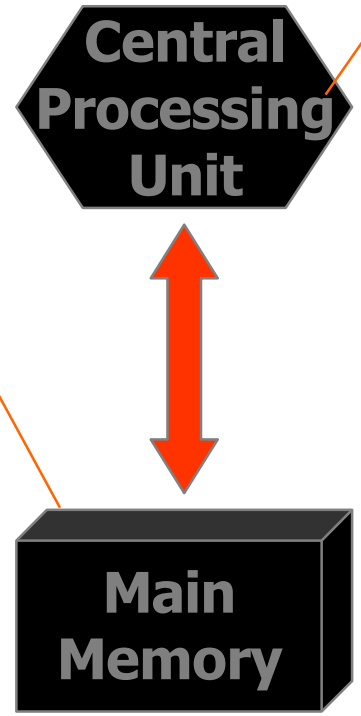
- **Software: Operating System, Application Programs**

- **UNIX is the name of a popular operating system.**

Typical Personal Computer System



Hardware Key Components (1) : CPU and Main Memory



Chip that executes program commands

Intel Core i7-13700K
Apple M3 Pro

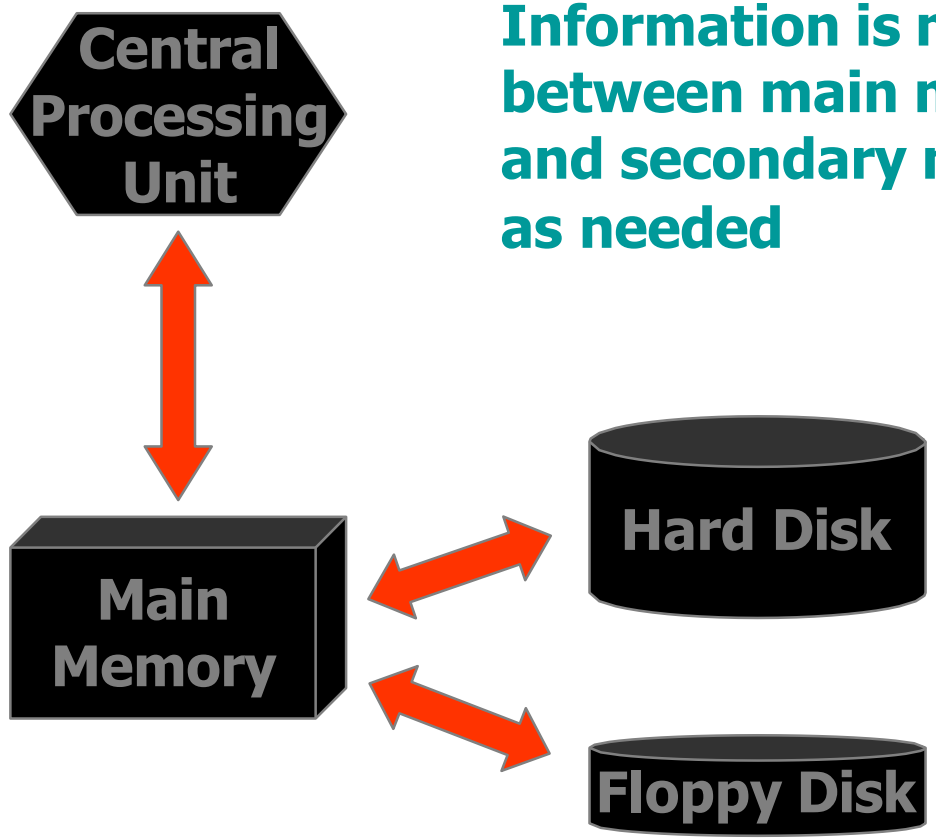
Primary storage area for programs and data that are in active use

Synonymous with RAM

Hardware Key Components (2) : Secondary Memory Devices

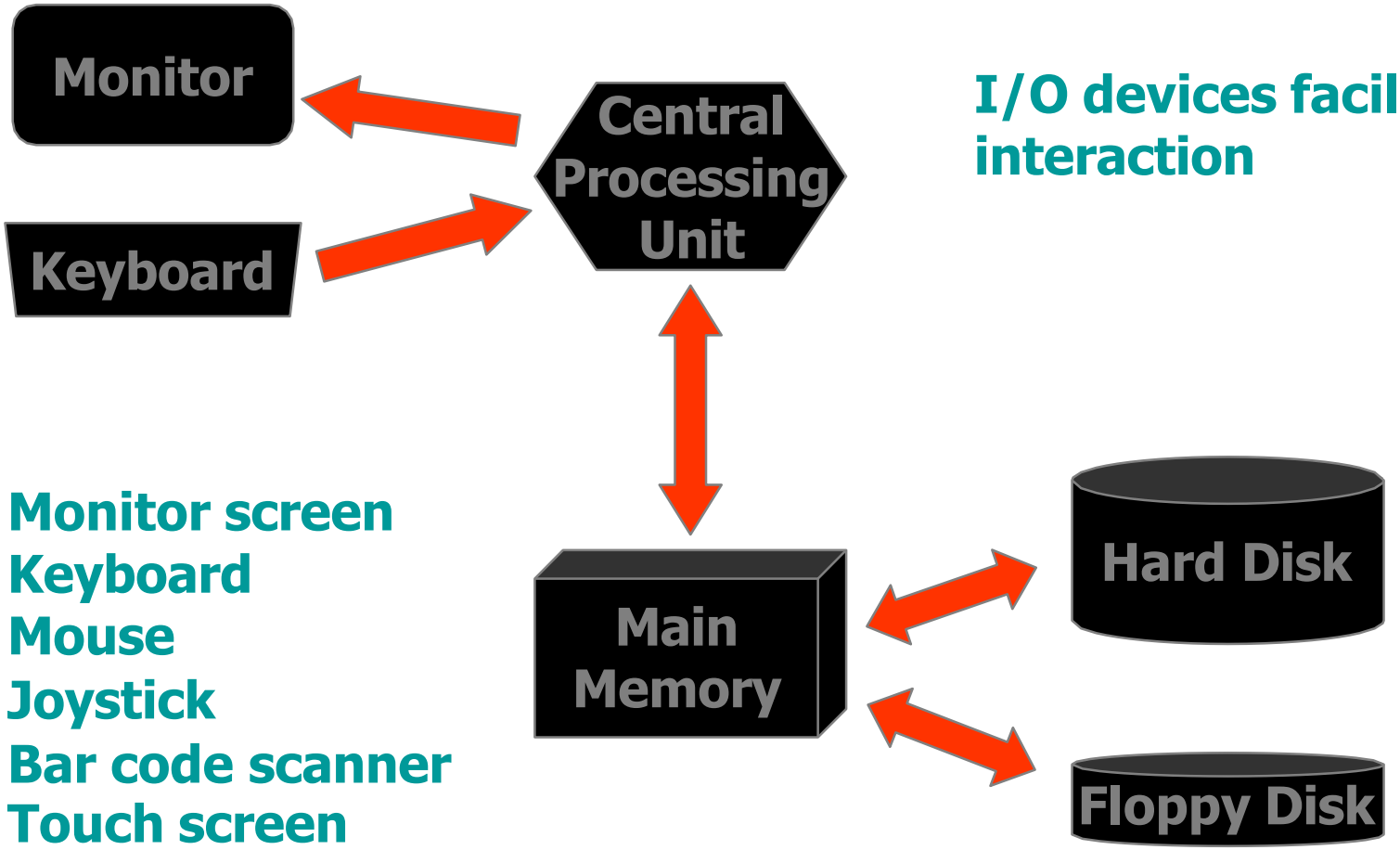
Secondary memory devices provide long-term storage

Information is moved between main memory and secondary memory as needed



Hard disks
Floppy disks
ZIP disks
Writable CDs
Tapes

Hardware Key Components (3) : Input / Output Devices

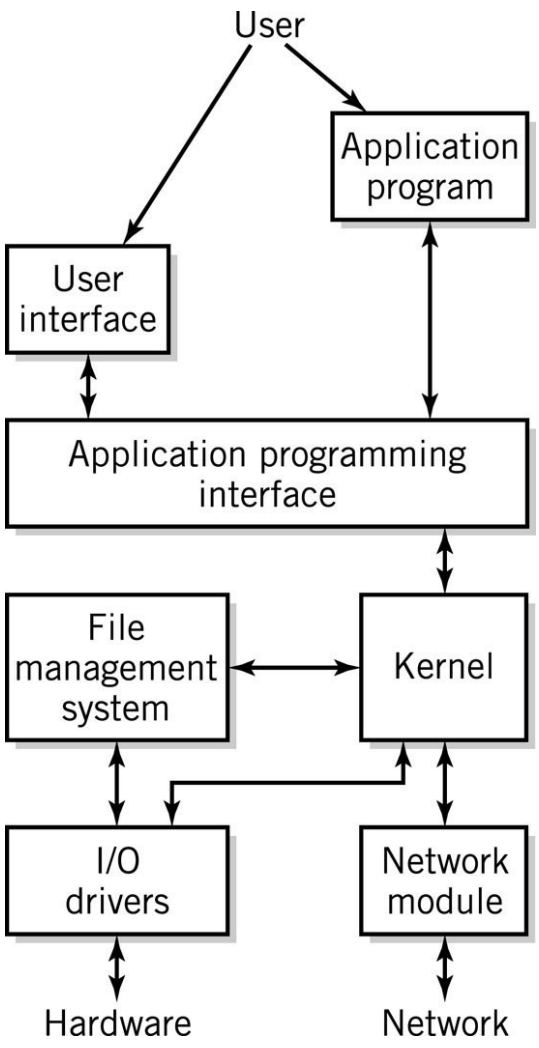


I/O devices facilitate use interaction

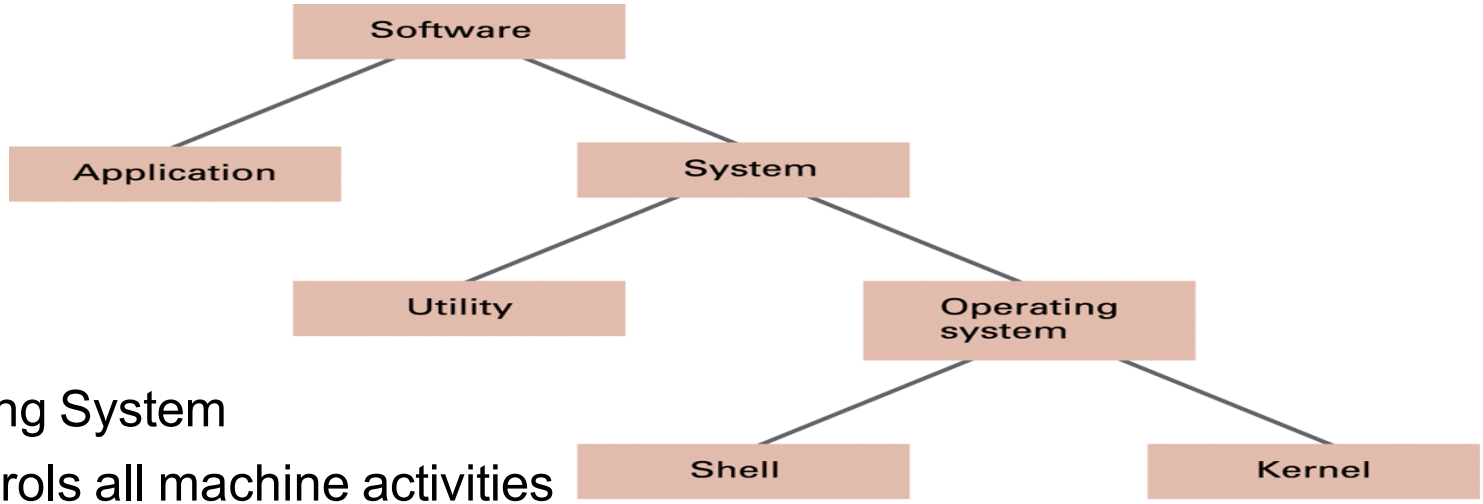
Monitor screen
Keyboard
Mouse
Joystick
Bar code scanner
Touch screen

Software Component

- Applications
- *Operating System*
  - API: application program interface
  - File management
  - I/O
  - Kernel
    - Memory management
    - Resource scheduling
    - Program communication
    - Security
  - Network Module



Software Categories



- Operating System
  - controls all machine activities
  - Oversee operation of computer - Store and retrieve files
  - Schedule programs for execution - Coordinate the execution of programs
  - Provides the user interface to the computer
  - manages resources such as the CPU and memory
- Application program
  - generic term for any other kind of software
  - word processors, missile control systems, games
- Most operating systems and application programs have a *graphical user interface* (GUI)

The shell as an interface between users and the operating system

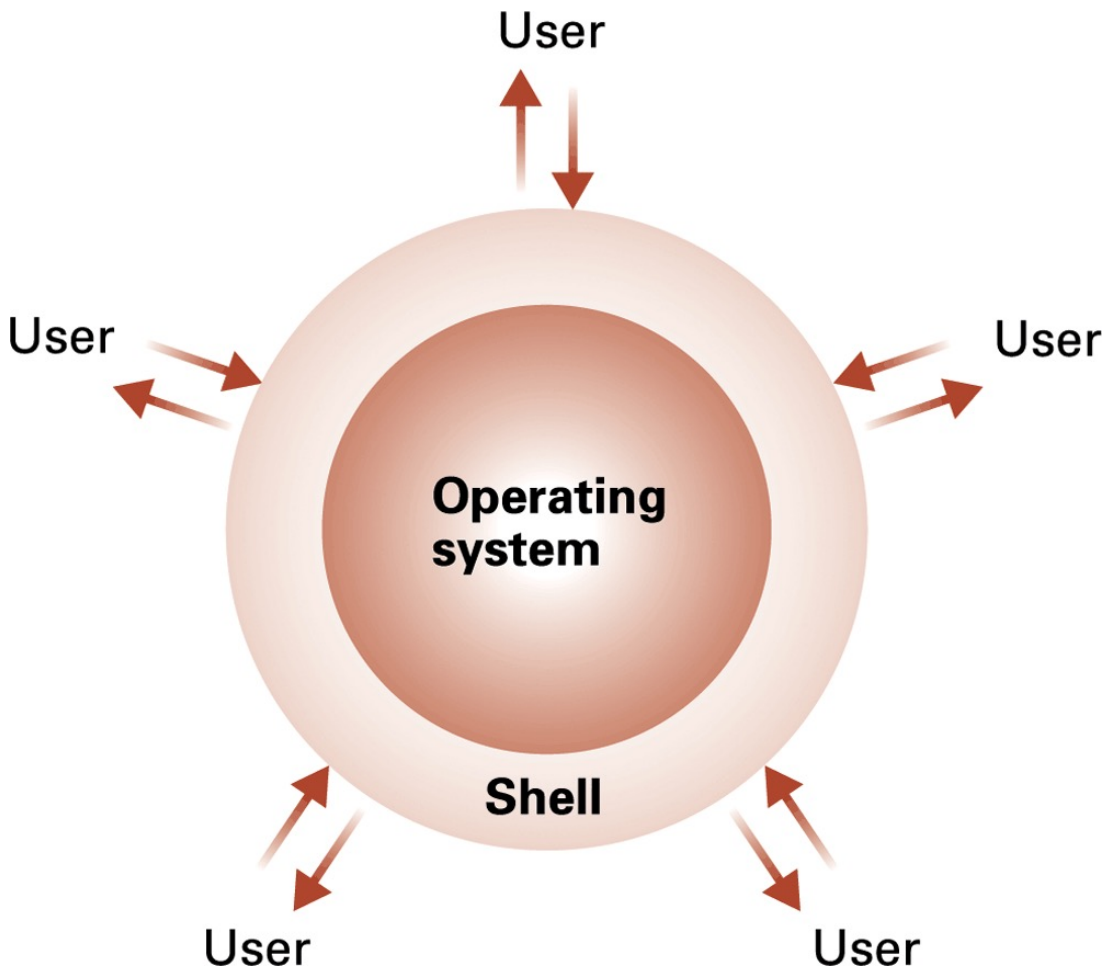
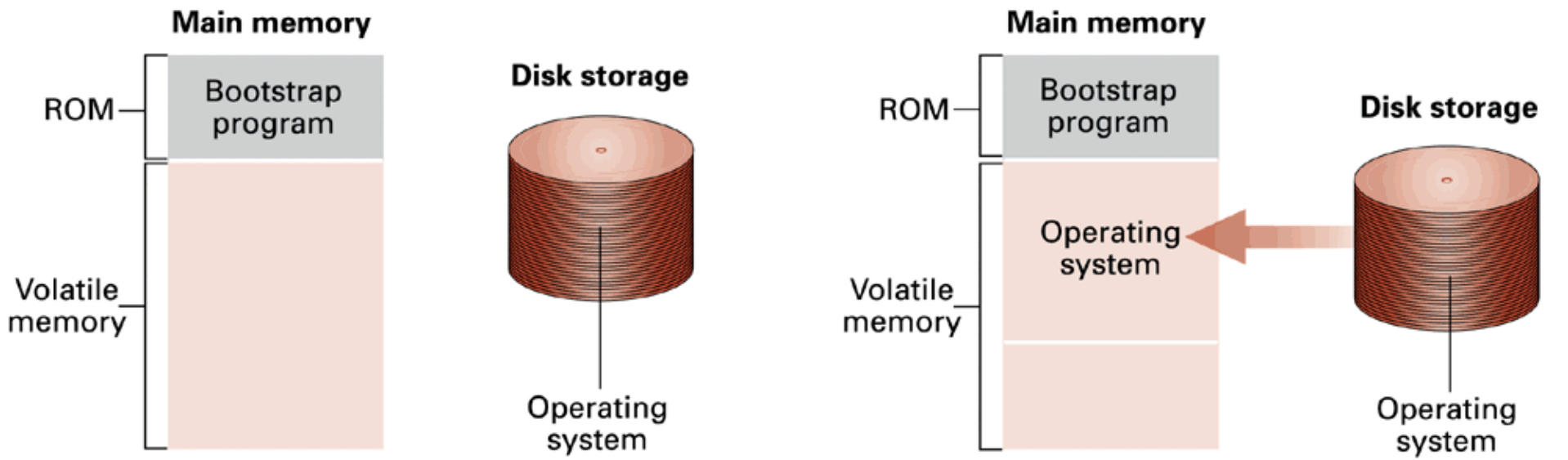


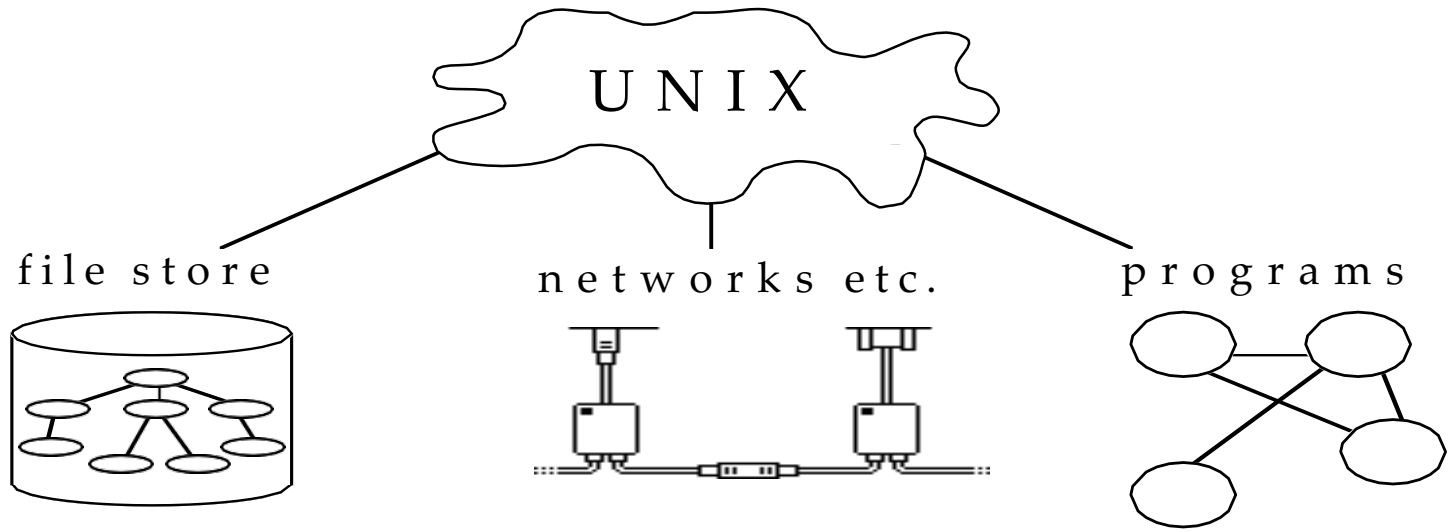
Figure 3.5 The booting process



Step 1: Machine starts by executing the bootstrap program already in memory. Operating system is stored in mass storage.

Step 2: Bootstrap program directs the transfer of the operating system into main memory and then transfers control to it.

UNIX is an operating system

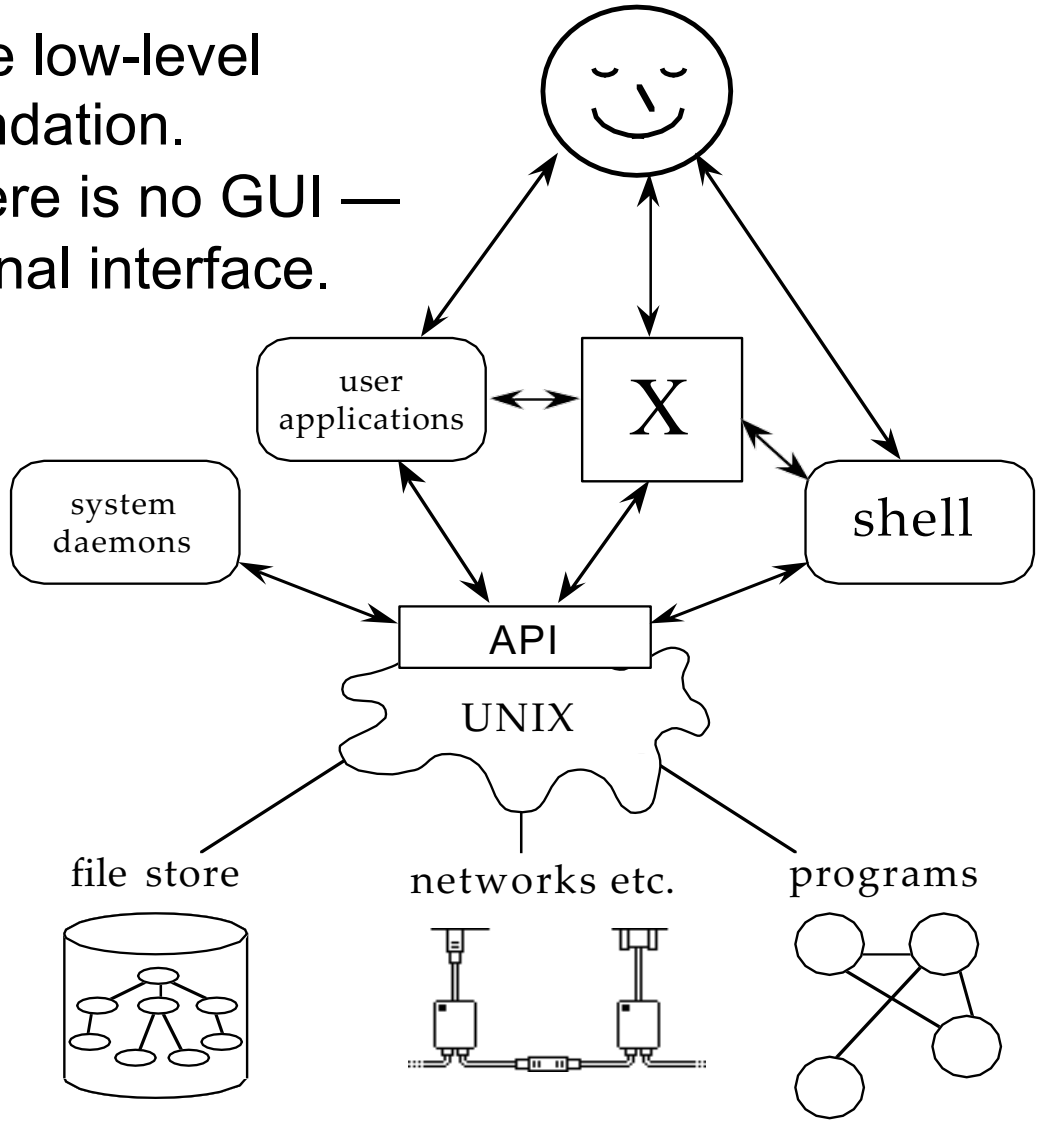


- **Provides a framework for executing programs and storing files**
  - **File: collection of data normally stored on disk.**
  - **Program: collection of instructions/data that is stored in a file.**

UNIX API – the system calls

ultimately everything works through system calls

X provides the low-level graphical foundation.
Without X, there is no GUI — only the terminal interface.



System Daemons

A *daemon* is a program with a unique purpose. They are utility programs that run silently in the background to monitor and take care of certain subsystems.

- printer daemon monitors printing services
- network daemon for network communications

Example:

httpd – the HTTP service manager. This is normally run with Web server software such as Apache.
sshd – Daemon responsible for managing the SSH service. This is used on virtually any server that accepts SSH connections.
(note the '**d**' at the end of system programs)
Use 'pstree' command and identify some daemons.

Process

- **When a program is executed, it is loaded into memory. It is called a *process* when it is executing.**
- **Most processes read/write data from/to files**
- **Processes and files have an owner**
- **UNIX supports hierarchical directory structure**
- **Files and processes have a location within the directory structure**
- **UNIX provides the capabilities to create, modify and destroy files, programs, and processes.**

Unix Attributes

- **Sharing of resources: CPU (time slices), memory (pages), disk (blocks)**

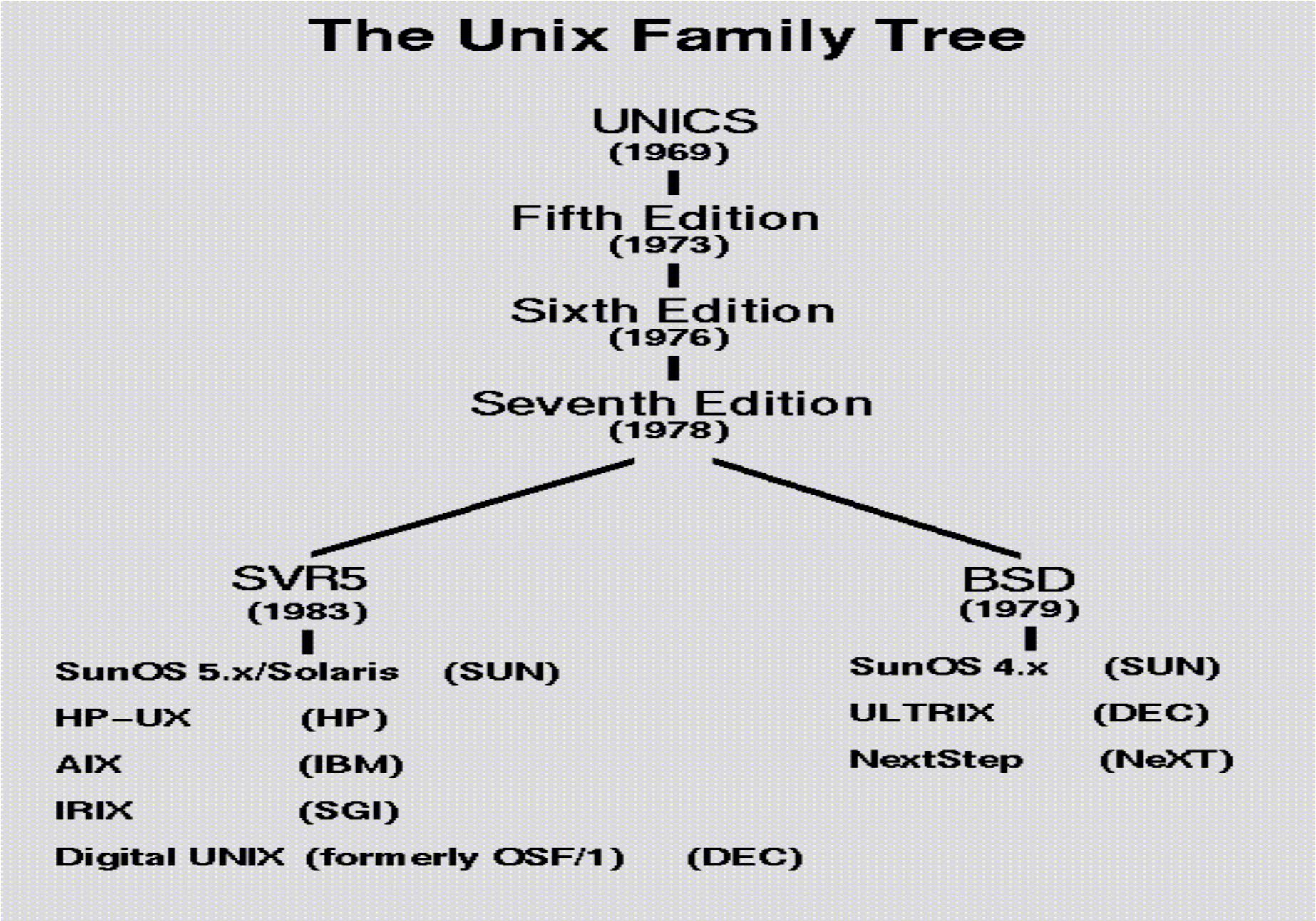
- **Communication: process-device controller, process-process, etc. (pipes 1-way, sockets 2-way)**

- **Utilities: Unix comes with a large collection of utilities; we will study many of these.**

- **Programmer support: Unix provides standard compilers for common system programming languages (like C and C++); Access to parallel processing, file handling and interprocess communication via System calls in C**

Advantages of UNIX

- It is **multitasking;** therefore, multiple programs can run at one time.
- It is **multiuser**, allowing more than a single user to work at any given time. This is accomplished by sharing processing time between each user and utilizing distributed computing systems.
- It is **safe**, preventing one program from accessing memory or storage space allocated to another, and enables protection.



Unics: Uniplexed Information and Computing Service, originally it was written in Assembly language and then in 1973 in C.

Two Main Varieties of Unix

- System V (AT&T) and
- BSD (Berkeley Standard Distribution)
  - Both are merged now. SunOS, IRIX, AIX, HP-UX have features from both varieties although most are System V Unix.
- Other Unix versions you may have heard of:
  - Linux (Fedora, Red Hat, Ubuntu, SUSE, etc.)
  - Sun Java Desktop OS, Solaris
  - Apple OS/X

Sub-Varieties of Unix

- Linux (Runs on PC architecture)
  - Fedora
  - Red Hat
  - Ubuntu
  - Sun Java Desktop OS, etc.
- Sun Solaris (Runs on SPARC architecture)
- Apple OS/X (Runs on PowerPC and Intel platforms)

Philosophies of Unix

- **Pipe mechanism**
  - **Output of one process can be used as input for another process. e.g.**
    - **$ who | sort**
  - **Using the pipe mechanism, complex tasks can be broken down into simpler ones and combined using pipes etc.**
- **Super user**
  - **user who has complete control over the system resources. Typically the System's Administrator.**

cat command and pipes

Pipes relieve the need for creating intermediate result files. See example in the screenshot:
cat command displays the contents of a file, and the *head* command is displaying first 3 lines of the *cat* command's output as a single concatenated operation

```
(base) Mds-MacBook-Pro:CSC 3320 System Programming mdmahfuzurrahman$ cat unix.txt
unix is great os. unix was developed in Bell labs.

learn operating system.

Unix linux which one you choose.

uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
(base) Mds-MacBook-Pro:CSC 3320 System Programming mdmahfuzurrahman$ cat unix.txt | head -3
unix is great os. unix was developed in Bell labs.

learn operating system.
(base) Mds-MacBook-Pro:CSC 3320 System Programming mdmahfuzurrahman$
```

script command

- *script* command is part of the *util-linux* package.
- To start recording the terminal session, we simply type *script* followed by the output file name
- To end the session recording, we simply type *exit*
- check our output.txt file using *cat*

Recording Terminal Session

```
[mrahman21@gsuad.gsu.edu@snowball ~]$ script mrahman_output.txt
Script started, file is mrahman_output.txt
[mrahman21@gsuad.gsu.edu@snowball ~]$ ls
hello  hello.c  mrahman_output.txt  sysprog.txt  unix.txt
[mrahman21@gsuad.gsu.edu@snowball ~]$ pwd
/home/mrahman21
[mrahman21@gsuad.gsu.edu@snowball ~]$ cat unix.txt
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uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
[mrahman21@gsuad.gsu.edu@snowball ~]$ exit
exit
Script done, file is mrahman_output.txt
```

Displaying Terminal Session

```
[mrahman21@gsuad.gsu.edu@snowball ~]$ cat mrahman_output.txt
Script started on Mon 19 Aug 2024 11:09:05 PM EDT
[mrahman21@gsuad.gsu.edu@snowball ~]$ ls
hello  hello.c  mrahman_output.txt  sysprog.txt  unix.txt
[mrahman21@gsuad.gsu.edu@snowball ~]$ pwd
/home/mrahman21
[mrahman21@gsuad.gsu.edu@snowball ~]$ cat unix.txt
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Unix linux which one you choose.

uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
[mrahman21@gsuad.gsu.edu@snowball ~]$ exit
exit

Script done on Mon 19 Aug 2024 11:09:30 PM EDT
[mrahman21@gsuad.gsu.edu@snowball ~]$
```